PROJECT TECHNICAL DOCUMENTATION

BioSpecInfo

Web platform for computational chemistry, physics and mathematics

Technical Dossier

DOCUMENT CONTROL

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BioSpecInfo — Technical Dossier

Open-access web platform for computational chemistry and the interactive consultation of chemical data on any device.

Author	Samuele Pio Provenzano
Thesis advisor	Prof. Savino Longo — University of Bari Aldo Moro
Profile	Cheminformatics Engineer / Scientific Software Developer
Software type	Progressive Web App (PWA), client-side, local-first, offline-first
Live demo	https://samupropio1-ship-it.github.io/BioSpecInfo-v11/
Repository	github.com/samupropio1-ship-it/BioSpecInfo-v11

> This dossier summarises the application and is the cover of the full technical documentation (SAD, Validation & Benchmark, Security & Compliance, Licensing) attached below.

1. Executive summary

BioSpecInfo is an **interactive cheminformatics platform** that runs all of its scientific logic **in the client browser**, with no server: molecular analysis, spectral prediction, 2D/3D modelling and educational tools — all **free, installable as an app and working offline**.

Its distinctive value is the **rare combination of advanced chemistry and modern software engineering** (WebAssembly, WebGL, PWA), with spectral predictors **quantitatively validated** against the official SDBS and NIST databases.

2. Product factsheet

Value proposition — A **≈4.5 MB** web platform (main file) that runs molecular analysis, spectral prediction and 3D visualisation at **zero server cost** and **100% offline-ready**.

Technical architecture	Content & capabilities
Client-side / Local-First (no backend)	**RDKit** cheminformatics engine (WebAssembly)
Single-file HTML5 / ES6, installable as **PWA**	**Spectra Centre**: IR, $^{1}H/^{13}C$-NMR, MS + isotope pattern, UV-Vis, Raman
2D (SVG/Canvas) and **3D** (three.js, 3Dmol.js) rendering	**Organic Chemistry**: 25 interactive modules
Service Worker offline-first, versioned	**Retrosynthesis**: 46 strategies + animated 2D/3D mechanisms

Technical architecture	Content & capabilities
Zero copyleft dependencies (MIT/BSD-3 only)	**3D Drug Atlas**: 63 diseases (23 tumours), 3D organs, PubMed
Local storage (localStorage, SQLite via sql.js)	3D astrochemistry, Quantum Lab, Academy (SM-2 quizzes)

Quantitative validation (thesis benchmark)

- **^{1}H-NMR** vs SDBS → **MAE 0.31 ppm** (89% of signals within ±0.5 ppm)
- **IR (C=O band)** vs NIST WebBook → **MAE 11 cm^{-1}** (90% within ±30 cm^{-1})
- Interactive performance **30–60 fps** from iPhone/iPad to Raspberry Pi 4

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3. Verified capabilities (current code state)

Spectra Centre (molecular editor & analysis) — draw from scratch (full periodic table, templates, drag-to-bond) or load via SMILES/library →

- All chemical notations: formula, canonical SMILES, CXSMILES, SMARTS, InChI,

InChIKey, mol block, common/IUPAC name, CAS (one-click copy).

- Predicted spectra with bibliographic sources; **isotope pattern** (IUPAC

abundances) and MS **adduct masses**.

- Functional groups and properties (H donors/acceptors, aromatic, rotatable,

stereocentres) **highlightable on the 2D and 3D structure**.

- Drug-likeness: **Lipinski, Veber, Egan, Ghose, Muegge**, ESOL solubility,

QED, **Brenk/PAINS structural alerts**, physicochemical radar.

- **Tanimoto** similarity (Morgan ECFP4 fingerprint), elemental analysis, SMARTS

substructure search, SVG/PNG/MOL export.

Synthesis & Retrosynthesis — 46 strategies with an **animated mechanism player** in **2D and 3D**: *electron-pushing* arrows, reaction intermediates, and a retrosynthesis mode showing **synthon formation**.

Organic Chemistry — 25 modules (fundamentals → heterocycles, pericyclic & Woodward–Hoffmann, organometallics & cross-coupling, radicals, spectroscopy, retrosynthesis & protecting groups).

Data Science — 3D Drug Atlas (13 body regions, 63 diseases of which 23 tumours, 3D organ models, PubMed links), ML labs, SQL playground.

Other — 3D astrochemistry, Quantum Lab, Academy with spaced repetition (SM-2), interactive simulations, databases of elements/drugs/biomolecules.

> Overall metrics declared in the thesis (historical data assets): 324 organic reactions in vector SVG graphics, 26 biosynthetic pathways, 140+ drugs, 80+ biomolecules, 118 elements, 10 quantum simulators.

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4. Technology stack

`HTML5` · `ES6/ESNext` · `WebAssembly` (RDKit MinimalLib, sql.js/SQLite) · `WebGL` (three.js, 3Dmol.js) · `SVG` / `Canvas 2D` · `Service Worker` / `PWA` · public API integration (PubChem PUG-REST, NIST, SDBS, MassBank, nmrshiftdb2, NASA/ESA).

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5. Author's declaration (self-attestation)

The undersigned **Samuele Pio Provenzano** declares that:

1. The software **BioSpecInfo** is the author's **original work**, except for the

third-party open-source libraries (RDKit, 3Dmol.js, three.js, SmilesDrawer, sql.js) used under their respective permissive licenses and documented in the *Open-Source Licensing & IP Matrix*.

1. The reported validation data (MAE 0.31 ppm for ^{1}H-NMR vs SDBS; MAE 11 cm^{-1} for

the IR C=O band vs NIST) were measured by the author as part of the thesis work; the predictors are **heuristic group-additivity** models, suitable for educational use and qualitative functional-group identification, with the limitations stated in the *Validation Report*.

1. The architecture is **local-first**: no user personal data is transmitted to

author-controlled servers (alignment with GDPR principles).

> This is an **author's declaration**, not a certification issued by a third party.

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6. Attached documentation

1. **Software Architecture Document (SAD)** — architecture, components, technologies.
2. **Verification & Validation & Benchmark Report** — test method, benchmarks, limitations.
3. **Security, Privacy & Compliance** — local-first, GDPR, OWASP, GAMP 5 (alignment).
4. **Open-Source Licensing & IP Matrix** — dependencies, licenses, intellectual property.

Software license: proprietary — "All rights reserved" (see `LICENSE`).